Notes: Off-policy strategy

The on-policy algorithm (`on_policy_mc_control`, see `rl_course/on_policy_monte_carlo/opmc_c/Section_4_on_policy_constant_alpha_mc.ipynb`) uses the same ϵ-greedy policy both to *generate* the episodes (the **behavior policy**) and to be *improved* (the **target policy**). This keeps things simple but forces a compromise — the policy must keep exploring ($\epsilon > 0$) even while we'd ultimately like it to be greedy/deterministic.

Off-policy methods break this coupling by using two distinct policies:

- **Behavior policy** $b(a|s)$: generates the experience (the actions actually taken in the environment). It must keep exploring, so it's typically ϵ-soft (e.g. ϵ-greedy).
- **Target policy** $\pi(a|s)$: the policy we are actually trying to learn/evaluate. It can be fully greedy with respect to $Q(s, a)$, since it never has to be executed during data collection.

This requires b to have **coverage** of π: whenever $\pi(a|s) > 0$, we need $b(a|s) > 0$ as well, otherwise we could never observe the actions π would take.

Why we need importance sampling

Returns G_t collected under b are distributed according to b, not π. To use them to estimate $Q_\pi(s, a)$, each return has to be reweighted by the **importance-sampling ratio**, i.e. the relative probability of the trajectory under π vs. under b:

$$\rho_{t:T-1} = \prod_{k=t}^{T-1} \frac{\pi(A_k \mid S_k)}{b(A_k \mid S_k)} \tag{1}$$

Two common estimators built from this ratio:

- **Ordinary importance sampling**: $Q(s, a) = \frac{1}{N} \sum_i \rho_i G_i$ — unbiased, but can have very high (even unbounded) variance since ρ is a product of many ratios.
- **Weighted importance sampling**: $Q(s, a) = \frac{\sum_i \rho_i G_i}{\sum_i \rho_i}$ — biased (bias vanishes asymptotically), but much lower variance in practice, so it's generally preferred.

A common special case: greedy target policy

If π is chosen to be the **greedy** policy w.r.t. Q (deterministic), then $\pi(A_k|S_k)$ is either 1 (the greedy action was taken) or 0 (a different, exploratory action was taken). This means:

- Any time the behavior policy takes a non-greedy (exploratory) action, ρ becomes 0 for the rest of that episode's prefix, and the trailing part of the trajectory can be discarded for the update.

- This is exactly the structure exploited by algorithms like **Q-learning**, which can be seen as an (incremental, bootstrapped) off-policy method with an implicit greedy target policy — no explicit importance-sampling ratio is needed there because the TD target already uses $\max_a Q(s', a)$.

On-policy vs. off-policy at a glance

	On-policy	**Off-policy**
Behavior policy	Same as target	Different from target (must have coverage)
Target policy	Must stay exploratory (ϵ-soft)	Can be fully greedy/deterministic
Sample reweighting	Not needed	Needed (importance sampling)
Variance	Lower	Higher (especially ordinary IS)
Data reuse	Harder — policy keeps changing	Easier — can learn from old/other agents' data, human demonstrations, replay buffers

Off-policy learning is what makes it possible to learn a target policy from data generated by a *different* source entirely (an older version of the policy, a hand-coded exploratory policy, logged human data, etc.), which is central to methods like **Q-learning**, **Expected SARSA** (off-policy variant), and off-policy **actor-critic** / **deep RL** algorithms (e.g. DQN, using a replay buffer of past experience).

Resources

[1] Reinforcement Learning: An Introduction. Ch. 5: Monte Carlo Methods